SANTIAGO CADAVID

Biomedical Engineer

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PROFILE

As Regional Manager at BV Medical, I combine my background in Biomedical Engineering and Technology Management to lead technical teams across the South Island. My role focuses on ensuring the safe, reliable, and compliant performance of medical equipment in healthcare environments while supporting healthcare professionals through equipment training and technical guidance.

EXPERIENCE

Regional manager / Biomedical engineer

BV Medical

October 2020 – Currently Christchurch, NZ

- Ensure company policies and procedures are followed.
- Implement company strategies at the regional level.
- Hire, train, and supervise technicians/engineers.
- Maintain relationships with key clients or partners.
- Communicate regional needs to head office.
- Performs preventive/scheduled maintenance on medical devices, and installs, calibrates, repairs and inspects medical equipment.
- Performs equipment repairs of substantial difficulty to a level requiring generic test instrumentation, or diagnostic software.
- Determines the need to remove and/or replace malfunctioning medical devices from service and implements accordingly.

Assistant manager

MK RD Limited

August 2018 – October 2020 Auckland, NZ

Field service engineer

Iforware S.A.S.

January 2017 – June 2017 Medellín, Colombia

- To design and create prototypes using open-source electronic prototyping platforms (i.e. Arduino) and commercial electronics.
- To operate with a variety of analog and digital transducers (temperature, humidity, dew point, gas concentration) for industrial and healthcare applications.
- To design and develop printed circuit boards (PCB) and assemble them.
- To develop hardware of traceability, telemetry and control.
- To support software tests in traceability, telemetry and control.
- To keep the correct operation of sensors and the telemetry system that were installed in hospitals and laboratories.

EDUCATION

PGDipSc in Science - Medical Physics

University of Canterbury

2024 – 2026

BSc Biomedical Engineering

Instituto Tecnológico Metropolitano - ITM

2012 – 2017

Diploma level 7 in Technology Management

Aspire2 International (New Zealand)

2017 – 2018

LANGUAGES

English Spanish

SKILLS

Medical Devices Electrical/Electronics Digital signal processing Medical Imaging Electrical Appliance Serviceperson (EAS)

Microsoft Office MATLAB LaTeX/Overleaf Arduino

Monitoring Equipment
Diagnostic Equipment
Life-support Equipment
Sterilization
Rehabilitation/Assistive E.
Radiology Equipment
Dental Equipment
Laboratory Equipment

RESEARCH EXPERIENCE

Research Fellow

Instituto Tecnológico Metropolitano - ITM

January 2016 – December 2016 Medellín, Colombia

- To evaluate different kinds of electrodes and their position configurations to improve the signal-noise ratio during surface electromyography (sEMG) signal acquisition.
- Acquisition of sEMG signals in healthy people during motor gestures involved during swallowing.
- To evaluate the activation patterns of each motor gesture performed.
- To prepare and carry out a workshop for students from different universities about electromyography and its applications.

PUBLICATIONS

Journal Articles

- Cadavid-Arboleda, Santiago et al. (2017). "**Assessment of Surface Electromyography During Orofacial Praxis in Healthy Subjects**". In: *VII Latin American Congress on Biomedical Engineering CLAIB 2016, Bucaramanga, Santander, Colombia, October 26th -28th, 2016* 60, pp. 165–168. DOI: 10.1007/978-981-10-4086-3. URL: http://link.springer.com/10.1007/978-981-10-4086-3.
- Cantillo-Mackenzie, German et al. (2017). "**Surface Electromyographic Characterization of Five Orofacial Ideomotor Praxis in 20 Healthy Individuals**". In: *VII Latin American Congress on Biomedical Engineering CLAIB 2016, Bucaramanga, Santander, Colombia, October 26th -28th, 2016* 60, pp. 221–224. DOI: 10.1007/978-981-10-4086-3. URL: http://link.springer.com/10.1007/978-981-10-4086-3.
- Restrepo-Agudelo, Sebastian et al. (2017). "**Improving surface EMG burst detection in infrahyoid muscles during swallowing using digital filters and discrete wavelet analysis**". In: *Journal of Electromyography and Kinesiology* 35, pp. 1–8. ISSN: 10506411. DOI: 10.1016/j.jelekin.2017.05.001. URL: http://www.sciencedirect.com/science/article/pii/S1050641116302991.

ADDITIONAL INFORMATION

- Electrical Appliance Serviceperson (EAS) / Credential ID EW154510

REFERENCES

References available upon request.